

# DAA Metacyc

## 0.1 Differential abundance analysis

- Group variable: **diet**
- Formula used:  $\sim \text{diet} * \text{time} + (1 | \text{id})$

Table 1: Table of significant features

| feature     | metadata | value  | coef    | qval_individual | qval_joint | model      |
|-------------|----------|--------|---------|-----------------|------------|------------|
| X5.1.99.... | time     | time   | -10.602 | 0.000           | 0.000      | prevalence |
| X5.1.99.... | diet     | 2      | -20.311 | 0.001           | 0.001      | prevalence |
| X5.1.99.... | diet     | 2:time | 10.603  | 0.002           | 0.003      | prevalence |
| X4.2.2.2    | diet     | 2:time | -2.360  | 0.007           | 0.009      | abundance  |
| X2.3.1.2... | diet     | 2:time | -0.778  | 0.021           | 0.018      | abundance  |
| X2.3.3.1... | diet     | 2      | 6.922   | 0.021           | 0.023      | abundance  |
| X1.1.1.1... | time     | time   | 10.079  | 0.021           | 0.018      | prevalence |
| X3.6.3.5    | diet     | 2      | -2.369  | 0.038           | 0.027      | abundance  |
| X3.1.1.8... | time     | time   | -0.862  | 0.050           | 0.036      | abundance  |
| X4.2.2.2    | diet     | 2      | 3.165   | 0.058           | 0.059      | abundance  |
| X1.2.1.6... | time     | time   | -1.857  | 0.060           | 0.047      | abundance  |
| X4.2.1.1... | diet     | 2:time | -0.885  | 0.060           | 0.047      | abundance  |
| X3.1.1.7... | diet     | 2:time | -2.355  | 0.063           | 0.048      | abundance  |
| X2.1.1.1... | diet     | 2:time | -1.913  | 0.070           | 0.091      | abundance  |
| X2.3.3.1... | diet     | 2:time | -3.639  | 0.070           | 0.091      | abundance  |
| X3.6.3.5    | diet     | 2:time | 1.380   | 0.070           | 0.059      | abundance  |
| X4.2.1.1... | diet     | 2:time | 2.345   | 0.070           | 0.091      | abundance  |
| X2.1.1.1... | diet     | 2      | 3.104   | 0.070           | 0.091      | abundance  |
| X2.1.1.1... | diet     | 2:time | 10.416  | 0.070           | 0.091      | prevalence |
| X2.4.1.2... | diet     | 2      | -8.348  | 0.070           | 0.091      | prevalence |
| X2.8.3.1... | diet     | 2:time | 11.176  | 0.073           | 0.095      | prevalence |
| X2.3.1.2... | diet     | 2      | 1.017   | 0.103           | 0.091      | abundance  |
| X5.1.1.1... | diet     | 2      | -1.887  | 0.103           | 0.091      | abundance  |
| X2.3.3.1... | diet     | 2:time | 1.252   | 0.671           | 0.091      | prevalence |
| X2.8.3.1... | diet     | 2:time | -1.960  | 0.694           | 0.095      | abundance  |
| X2.3.3.1... | diet     | 2      | -1.731  | 0.759           | 0.023      | prevalence |
| X5.1.99.... | diet     | 2:time | -0.498  | 0.791           | 0.003      | abundance  |
| X5.1.99.... | diet     | 2      | 0.704   | 0.826           | 0.001      | abundance  |
| X2.1.1.1... | diet     | 2:time | 0.628   | 0.880           | 0.091      | abundance  |
| X4.2.2.2    | diet     | 2      | -3.784  | 0.898           | 0.059      | prevalence |
| X2.1.1.1... | diet     | 2      | -3.229  | 0.917           | 0.091      | prevalence |
| X5.1.99.... | time     | time   | -0.191  | 0.931           | 0.000      | abundance  |
| X4.2.2.2    | diet     | 2:time | 1.892   | 0.945           | 0.009      | prevalence |
| X2.1.1.1... | diet     | 2:time | 0.956   | 0.997           | 0.091      | prevalence |
| X2.4.1.2... | diet     | 2      | -0.076  | 1.000           | 0.091      | abundance  |
| X4.2.1.1... | diet     | 2:time | 9.941   | 1.000           | 0.091      | prevalence |
| X1.2.1.6... | time     | time   | 0.000   | 1.000           | 0.047      | prevalence |

Table 1: Table of significant features (*continued*)

| feature     | metadata | value  | coef   | qval_individual | qval_joint | model      |
|-------------|----------|--------|--------|-----------------|------------|------------|
| X1.1.1.1... | time     | time   | -1.281 | NA              | 0.018      | abundance  |
| X2.3.1.2... | diet     | 2      | NA     | NA              | 0.091      | prevalence |
| X2.3.1.2... | diet     | 2:time | NA     | NA              | 0.018      | prevalence |
| X3.1.1.7... | diet     | 2:time | 20.630 | NA              | 0.048      | prevalence |
| X3.1.1.8... | time     | time   | NA     | NA              | 0.036      | prevalence |
| X3.6.3.5    | diet     | 2      | NA     | NA              | 0.027      | prevalence |
| X3.6.3.5    | diet     | 2:time | NA     | NA              | 0.059      | prevalence |
| X4.2.1.1... | diet     | 2:time | NA     | NA              | 0.047      | prevalence |
| X5.1.1.1... | diet     | 2      | NA     | NA              | 0.091      | prevalence |









